

Huiles minérales isolantes
dans les matériels électriques –
Lignes directrices pour la maintenance
et la surveillance

Mineral insulating oils in electrical equipment –
Supervision and maintenance guidance

ISSUED BY *Blanchard*
(DIV. M.R.)
DATE 01/07/2008

CONTROLLED COPY
SR. NO.
VALID TILL NEXT CHANGE

CONTROLLED COPY
2
only a stamp is RED

Copyright International Electrotechnical Commission (IEC), Geneva, Switzerland. All rights reserved. This copy has been made by Book Supply Bureau under license from the IEC. No resale of this document is permitted. No part of this document may be copied or reproduced in any form by any means (graphic, electronic or mechanical), including photocopying, recording, recording taping or information and retrieval system), nor made available on the internet, a public network, by satellite or otherwise without the prior written consent of the IEC.



CEI
IEC
60422
Troisième édition
Third edition
2005-10

NORME
INTERNATIONALE
INTERNATIONAL
STANDARD

Huiles minérales isolantes
dans les matériels électriques –
Lignes directrices pour la maintenance
et la surveillance

Mineral insulating oils in electrical equipment –
Supervision and maintenance guidance

© IEC 2005 Droits de reproduction réservés — Copyright - all rights reserved

Aucune partie de cette publication ne peut être reproduite ni
utilisée sous quelque forme que ce soit et par aucun procédé,
électronique ou mécanique, y compris la photocopie et les
microfilms, sans l'accord écrit de l'éditeur.

No part of this publication may be reproduced or utilized in any
form or by any means, electronic or mechanical, including
photocopying and microfilm, without permission in writing from
the publisher.

International Electrotechnical Commission, 3, rue de Varembé, PO Box 131, CH-1211 Geneva 20, Switzerland
Telephone: +41 22 919 02 11 Telefax: +41 22 919 03 00 E-mail: inmail@iec.ch Web: www.iec.ch



Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

CODE PRIX
PRICE CODE

X

For price, see current catalogue
Pour prix, voir catalogue en vigueur

CONTENTS

FOREWORD	7
INTRODUCTION	11
1 Scope	13
2 Normative references	13
3 Terms and definitions	15
4 General remarks	17
4.1 General caution	17
4.2 Environment	17
5 Properties and deterioration/degradation of oil	19
6 Oil tests and their significance	19
6.1 General	19
6.2 Colour and appearance	21
6.3 Breakdown voltage	21
6.4 Water content	23
6.5 Acidity	29
6.6 Dielectric Dissipation Factor (DDF) and resistivity	29
6.7 Inhibitor content and oxidation stability	35
6.8 Sediment and sludge	37
6.9 Interfacial tension (IFT)	37
6.10 Particle count	37
6.11 Flash-point	39
6.12 Compatibility of insulating oils	39
6.13 Pour-point	41
6.14 Density	41
6.15 Viscosity	41
6.16 Polychlorinated biphenyls (PCBs)	41
6.17 Corrosive sulphur	41
7 Sampling of oil from equipment	43
8 Categories of equipment	43
9 Evaluation of mineral insulating oil in new equipment	45
10 Evaluation of oil in service	47
10.1 General	47
10.2 Frequency of examination of oils in service	47
10.3 Testing procedures	51
10.4 Classification of the condition of oils in service	51
10.5 Corrective action	53
11 Handling and storage	63
12 Treatment	65
12.1 General	65
12.2 Reconditioning	65
12.3 Reclaiming	73
12.4 Decontamination of oils containing PCB	75

13	Replacement of oil in electrical equipment.....	77
13.1	Replacement of oil in transformers rated below 72,5 kV and in switchgear and associated equipment.....	77
13.2	Replacement of oil in transformers rated 72,5 kV and above.....	77
13.3	Replacement of oil in electrical equipment contaminated with PCB.....	77
Annex A (informative)	Sampling temperature below 20 °C.....	79
Annex B (informative)	Particles.....	81
Annex C (informative)	Test method for determination of sediment and sludge.....	83
Bibliography.....		85
Figure 1 – Example of the variation in saturation water content with oil temperature and acidity for insulating oil originally conforming to IEC 60296.....		25
Figure 2 – Typical correction factors.....		29
Figure 3 – Example of variation of resistivity with temperature for insulating oils.....		33
Table 1 – Tests for mineral insulating oils.....		21
Table 2 – Categories of equipment.....		43
Table 3 – Recommended limits for mineral insulating oils after filling in new electrical equipment prior to energization.....		45
Table 4 – Recommended frequency of testing(1).....		49
Table 5 – Application and interpretation of tests.....		55
Table 6 – Corrective actions.....		63
Table 7 – Conditions for processing inhibited mineral insulating oils.....		67
Table A.1 – Guidelines for interpreting data expressed in percent saturation.....		79
Table B.1 – Examples of contamination levels (particles) encountered on power transformer insulating oil as measured by laser counting (ISO 4406).....		81

INTERNATIONAL ELECTROTECHNICAL COMMISSION

MINERAL INSULATING OILS IN ELECTRICAL EQUIPMENT –
SUPERVISION AND MAINTENANCE GUIDANCE

FOREWORD

1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.

2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.

3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.

4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.

5) IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with an IEC Publication.

6) All users should ensure that they have the latest edition of this publication.

7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.

8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.

9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International standard IEC 60422 has been prepared by IEC technical committee 10: Fluids for electrotechnical applications.

This third edition cancels and replaces the second edition, published in 1989, and constitutes a technical revision.

The main changes with regard to the previous edition are as follows:

This standard has been revised to take into account changes in oil and equipment technology and to have due regard for the best practices currently in use world-wide.

The action limits for all oil tests have been revised and changes made where necessary to enable users to use current methodology and comply with requirements and regulations affecting safety and environmental aspects.

This guidance incorporates changes introduced in associated standards since the publication of the second edition.

The text of this standard is based on the following documents:

FDIS	10/636/FDIS
Report on voting	10/641/RVD

Full information on the voting for the approval of this guide can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

INTRODUCTION

Insulating mineral oils are used in electrical equipment employed in the generation, transmission, distribution and use of electrical energy, so that the amount of oil in service, worldwide, amounts to hundreds of millions of kilograms.

Monitoring and maintaining oil quality is essential to ensure the reliable operation of oil-filled electrical equipment. Codes of practice for this purpose have been established by electrical power authorities, power companies and industries in many countries.

A review of current experience reveals a wide variation of procedures and criteria. It is possible, however, to compare the value and significance of standardized oil tests and to recommend uniform criteria for the evaluation of test data.

If a certain amount of oil deterioration (by degradation or contamination) is exceeded, there is inevitably some erosion of safety margins and the question of the risk of premature failure should then be considered. While the quantification of the risk can be very difficult, a first step involves the identification of potential effects of increased deterioration. The philosophy underlying this standard is to furnish users with as broad a base of understanding of oil quality deterioration as is available, so that they can make informed decisions on inspection and maintenance practices.

Unused mineral oils are limited resources and should be handled with this in mind. Used mineral oils are, by most regulations, deemed to be controlled waste. If spills occur, this may have a negative environmental impact, especially if the oil is contaminated by persistent organic pollutants such as polychlorinated biphenyls (PCB).

The guidelines given in this standard, whilst technically sound, are mainly intended to serve as a common basis for the preparation of more specific and complete codes of practice by users in the light of local circumstances. Sound engineering judgement will have to be exerted in seeking the best compromise between technical requirements and economic factors.

Reference should also be made to instructions from the equipment manufacturer.

MINERAL INSULATING OILS IN ELECTRICAL EQUIPMENT – SUPERVISION AND MAINTENANCE GUIDANCE

1 Scope

This International Standard gives guidance on the supervision and maintenance of the quality of the insulating oil in electrical equipment.

This standard is applicable to mineral insulating oils, originally supplied conforming to IEC 60296, and used in transformers, switchgear and other electrical apparatus where oil sampling is reasonably practicable and where the normal operating conditions specified in the equipment specifications apply.

This standard assists the power equipment operator to evaluate the condition of the oil and maintain it in a serviceable condition. It also provides a common basis for the preparation of more specific and complete local codes of practice.

This standard includes recommendations on tests and evaluation procedures and outlines methods for reconditioning and reclaiming oil and the decontamination of oil contaminated with PCB.

NOTE The condition monitoring of electrical equipment, for example by analysis of dissolved gases, furanic compounds or other means is outside the scope of this standard.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60156: *Insulating liquids – Determination of the breakdown voltage at power frequency – Test method*

IEC 60247: *Insulating liquids – Measurement of relative permittivity, dielectric dissipation factor (tan δ) and d.c. resistivity*

IEC 60296: *Fluids for electrotechnical applications – Unused mineral insulating oils for transformers and switchgear*

IEC 60475: *Method of sampling liquid dielectrics*

IEC 60666: *Detection and determination of specified anti-oxidant additives in insulating oils*

IEC 60814: *Insulating liquids – Oil-impregnated paper and pressboard – Determination of water by automatic coulometric Karl Fischer titration*

IEC 60970: *Methods for counting and sizing particles in insulating liquids*

IEC 61125: *Unused hydrocarbon-based insulating liquids – Test methods for evaluating the oxidation stability*

IEC 61619: *Insulating liquids – Contamination by polychlorinated biphenyls (PCBs) – Method of determination by capillary column gas chromatography*

IEC 62021-1: *Insulating liquids – Determination of acidity – Part 1: Automatic potentiometric titration*

ISO 2049: *Petroleum products – Determination of colour (ASTM scale)*

ISO 2719: *Determination of flash point – Pensky-Martens closed cup method*

ISO 3016: *Petroleum products – Determination of pour point*

ISO 3104: *Petroleum products – Transparent and opaque liquids – Determination of kinematic viscosity and calculation of dynamic viscosity*

ISO 3675: *Crude petroleum and liquid petroleum products – Laboratory determination of density – Hydrometer method*

ASTM D971-99a: 2004 *Standard test method for interfacial tension of oil against water by the ring method*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1

distribution transformer

small transformer, usually less than 2 MVA, used by electrical utilities for the distribution of electricity

3.2

industrial or strategic transformer

transformer that, irrespective of its rated power and voltage, is used by industry in general and has a critical role in the proper functioning of such industry

3.3

local regulations

regulations pertinent to the particular process in the country concerned

3.4

routine tests (Group 1)

minimum tests required to monitor the oil and to ensure that it is suitable for continued service

NOTE If the results obtained from these tests do not exceed recommended action limits usually no further tests are considered necessary until the next regular period for inspection but, under certain perceived conditions, complementary tests may be deemed prudent.

3.5
complementary tests (Group 2)
 additional tests, which may be used to obtain further specific information about the quality of the oil, and may be used to assist in the evaluation of the oil for continued use in service

3.6
special investigative tests (Group 3)
 used mainly to determine the suitability of the oil for the type of equipment in use and to ensure compliance with environmental and operational considerations

3.7
reconditioning
 process that eliminates water and solid particles and contaminants by physical processing only

3.8
reclamation
 process that eliminates soluble and insoluble polar contaminants from the oil by chemical and physical processing

3.9
PCB decontamination
 process by which PCB contamination may be removed from mineral oil

4 General remarks

4.1 General caution

This International Standard does not purport to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate health and safety practices and determine the applicability of regulatory limitations prior to use.

The mineral oils which are the subject of this standard should be handled with due regard to personal hygiene. Direct contact with the eyes may cause slight irritation. In the case of eye contact, irrigation with copious quantities of clean running water should be carried out and medical advice sought. For more information, refer to the safety data sheet provided by the manufacturer. Some of the tests specified in this standard involve the use of processes that could lead to a hazardous situation. Attention is drawn to the relevant standard for guidance.

4.2 Environment

This standard is applicable to mineral oils, chemicals and used sample containers.

Attention is drawn to the fact that, at the time of writing this guide, some mineral oils in service are known to be contaminated to some degree by PCBs. Because of this, safety countermeasures shall be taken to avoid risks to workers, the public and the environment during the life of the equipment, by strictly controlling spills and emissions. Disposal or decontamination of these oils shall be carried out strictly in accordance with local regulations. Every precaution should be taken to prevent release of mineral oil into the environment.

5 Properties and deterioration/degradation of oil

The reliable performance of mineral insulating oil in an insulation system depends upon certain basic oil characteristics that can affect the overall performance of the electrical equipment.

In order to accomplish its multiple role of dielectric, coolant and arc-quencher, the oil needs to possess certain properties, in particular:

- high dielectric strength to withstand the electric stresses imposed in service;
- sufficiently low viscosity so that its ability to circulate and transfer heat is not impaired;
- adequate low-temperature properties down to the lowest temperature expected at the installation site;
- resistance to oxidation to maximise service life.

In service, mineral oil degrades due to the conditions of use. In many applications, insulating oil is in contact with air and is therefore subject to oxidation. Elevated temperatures accelerate degradation. The presence of metals, organo-metallic compounds or both may act as a catalyst for oxidation. Changes in colour, the formation of acidic compounds and, at an advanced stage of oxidation, precipitation of sludge, may occur. Dielectric and, in extreme cases, thermal properties may be impaired.

In addition to oxidation products, many other undesirable contaminants, such as water, solid particles and oil-soluble polar compounds can accumulate in the oil during service and affect its electrical properties.

The presence of such contaminants and any degradation product of oil are indicated by a change of one or more properties as described in Table 1.

Deterioration of other constructional materials, which may interfere with the proper functioning of the electrical equipment and shorten its working life, may also be indicated by changes in oil properties.

6 Oil tests and their significance

6.1 General

A large number of tests can be applied to mineral insulating oils in electrical equipment. The tests listed in Table 1 and discussed in 6.2 to 6.10 are considered sufficient to determine whether the condition of the oil is adequate for continued operation and to suggest the type of corrective action required, where applicable. The tests are not listed in order of priority.

Table 1 – Tests for mineral insulating oils

Property	Group ^d	Subclause	Method
Colour and appearance	1	6.2	ISO 2049
Breakdown voltage	1	6.3	IEC 60156
Water content	1	6.4	IEC 60814
Acidity (neutralization value)	1	6.5	IEC 62021-1
Dielectric dissipation factor (DDF) or resistivity	1	6.6	IEC 60247
Inhibitor content ^c	1	6.7	IEC 60666
Sediment and sludge	2	6.8	Annex C
Interfacial tension (IFT) ^a	2	6.9	ASTM D971-99a
Particles (particle count)	2	6.10	IEC 60970
Oxidation stability ^a	3	6.7.4	IEC 61125
Flash-point ^b	3	6.11	ISO 2719
Compatibility ^b	3	6.12	IEC 61125
Pour-point ^b	3	6.13	ISO 3016
Density ^b	3	6.14	ISO 3675
Viscosity ^b	3	6.15	ISO 3104
Polychlorinated biphenyls (PCBs)	3	6.16	IEC 61619
Corrosive sulphur ^a	3	6.17	DIN 51353 [5] ¹
^a Only needed under special circumstances, see applicable subclause. ^b Not essential, but can be used to establish type identification. ^c Restricted to inhibited oils. ^d Group 1 are routine tests, group 2 are complementary tests, group 3 are special investigative tests.			

6.2 Colour and appearance

The colour of an insulating oil is determined in transmitted light and is expressed by a numerical value based on comparison with a series of colour standards. It is not a critical property, but it may be useful for comparative evaluation. A rapidly increasing or a high colour number may be an indication of oil degradation or contamination.

Besides colour, the appearance of oil may show cloudiness or sediments, which may indicate the presence of free water, insoluble sludge, carbon, fibres, dirt, or other contaminants.

6.3 Breakdown voltage

Breakdown voltage is a measure of the ability of oil to withstand electric stress.

Dry and clean oil exhibits an inherently high breakdown voltage. Free water and solid particles, the latter particularly in combination with high levels of dissolved water, tend to migrate to regions of high electric stress and reduce breakdown voltage dramatically. The measurement of breakdown voltage, therefore, serves primarily to indicate the presence of contaminants such as water or particles. A low value of breakdown voltage can indicate that one or more of these are present. However, a high breakdown voltage does not necessarily indicate the absence of all contaminants.

¹ References in square brackets refer to the bibliography.

6.4 Water content

6.4.1 General

Depending on the amount of water, the temperature of the insulating system and the status of the oil, the water content of insulating oils influences:

- the breakdown voltage of the oil;
- the solid insulation and
- the ageing tendency of the liquid and solid insulation.

The water content in the liquid and solid insulation thus has a significant impact on the actual operating conditions and the lifetime of the transformer.

There are two main sources of water increase in transformer insulation:

- ingress of moisture from the atmosphere;
- degradation of cellulose and oil.

Water is transferred in oil filled electrical equipment by the insulating liquid. Water is present in oil in a dissolved form and may also be present as a hydrate adsorbed by polar ageing products (bonded water). Particles, such as cellulose fibres may bond some water.

6.4.2 Water in oil

The solubility of water in oil, given in mg/kg, depends on the condition of the oil, the temperature and type of oil. The absolute water content (W_{abs}) is independent of the temperature, type and condition of the oil and the result is given in mg/kg. W_{abs} can be measured according to IEC 60814. The relative water content (W_{rel}) is defined by the ratio W_{abs} /water solubility and the result is given in percent. Water solubility should be determined at the same temperature as the oil has been taken.

The water content in oil is directly proportional to the relative water concentration (relative saturation) up to the saturation level. The temperature dependence of the solubility of water in oil (W_s) is expressed by

$$W_s = W_{oil} e^{-B/T} \quad (1)$$

where T is the temperature of the oil at the point of sampling in Kelvin and W_{oil} and B are constants that are similar for many transformer oils but may be different for some products, mainly due to differences in aromatic contents. At elevated temperatures, some amount of hydrated water may transfer into dissolved water.

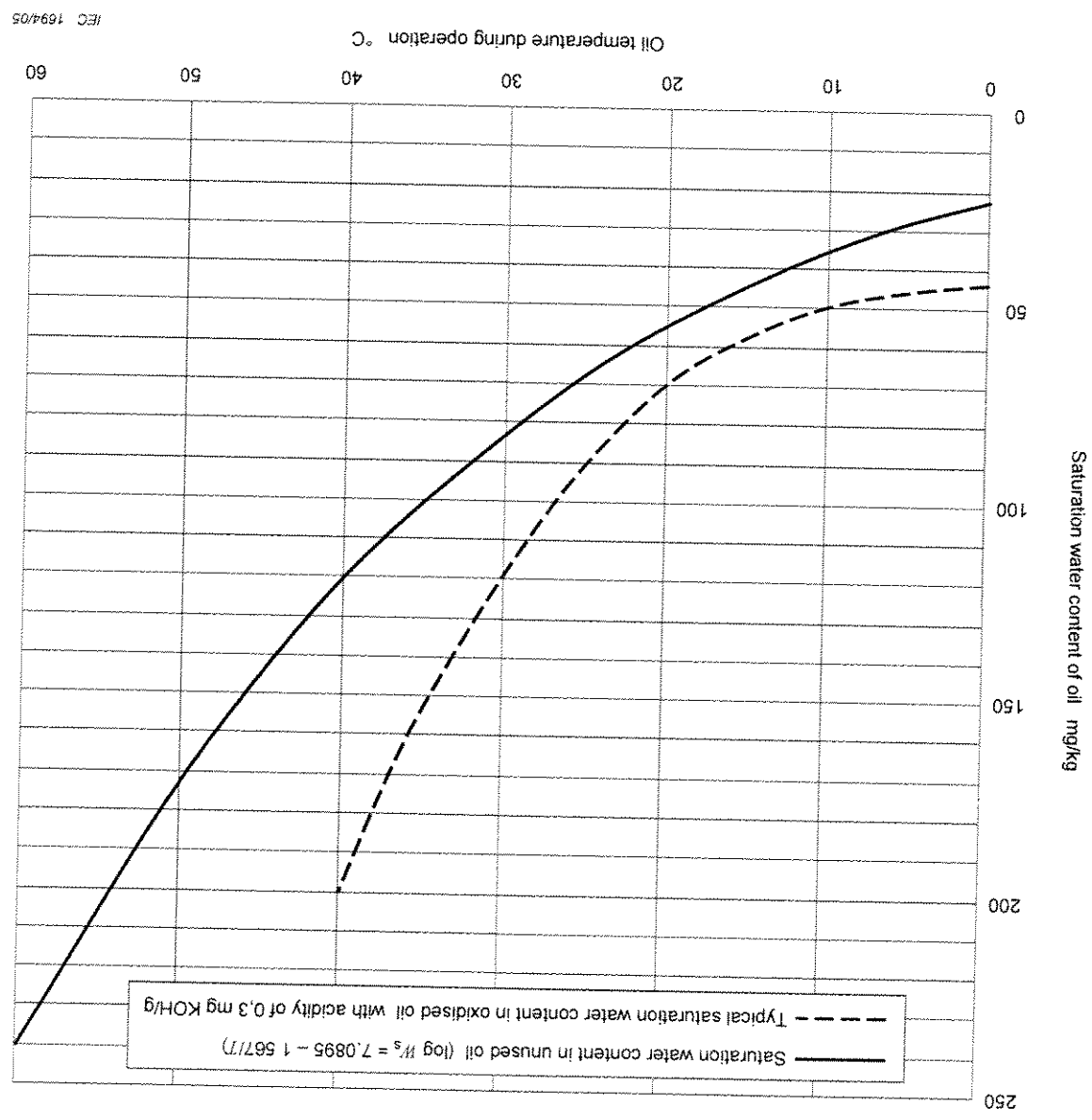
As oils become very oxidized with increasing amounts of polar ageing by-products, their water solubility characteristics, which are also dependent on the type of the oil, also increase. The solubility of water in very aged oils may be much higher than that in unused oils (Figure 1). Each oil should be considered separately and no universal formula is available.

Transformers are dried during the manufacturing process until measurements or standard practices would yield a moisture content in the cellulosic insulation of less than 0.5 % to 1.0 % depending upon purchaser's and manufacturer's requirements. After the initial drying, the moisture content of the insulation system increases depending on the environmental and/or operating conditions.

In a transformer, the total mass of water is distributed between the paper and the oil such that the bulk of water is in the paper. Small changes in temperature significantly modify the dissolved water content of the oil but only slightly modify the water content of the paper.

6.4.3 Water content in the oil/paper-system

Figure 1 – Example of the variation in saturation water content with oil temperature and acidity for insulating oil originally conforming to IEC 60296



When oil in a transformer is working at a constant, relatively elevated temperature for a long period, thermodynamic equilibrium between water absorbed by cellulose and water dissolved in oil is closely approached. This equilibrium is temperature-dependent so that at elevated temperatures more water diffuses from the paper into the oil. However, if the oil temperature is not high enough, such an equilibrium is not reached because of the lower rate of diffusion of water to the oil from the cellulose insulation.

The determination of the water content in the paper of a transformer by the measurement of the water in oil has been frequently described, but practical results are often not in line with the theoretical predictions. The drying process of the paper may not take out as much water as calculated.

All calculations and correlations of the water content in oil and the water content in the oil/paper-system depend on the equilibrium state between the insulating oil and the oil/paper-system and vice versa. Many factors such as the difference in the temperature between oil and the cellulose/oil-system influence the equilibrium. The calculation of the water content of the paper/pressboard by determination of water in the oil has been examined in several studies and publications [2, 4, 6].

6.4 Interpretation of results

6.4.1 General

The tools presented in this subclause for interpreting the results are applicable only if the following conditions are fulfilled:

- equilibrium exists between oil and paper;
- no abnormal ingress of water (leaks);
- presence of paper in the equipment;
- absence of free water.

6.4.2 Sampling temperature at or above 20 °C

For the proper interpretation of moisture content, the analytical result of water content of the oil at a given sampling temperature needs to be corrected to that at a defined temperature. For practical reasons, the defined temperature is set at 20 °C, since below 20 °C the rate of diffusion of water is too slow to achieve equilibrium in operational equipment

The correction formula, as demonstrated by several independent studies (Figure 2), is:

$$f = 2,24e^{(-0,04t_s)} \quad (2)$$

where f is the correction factor and t_s is the oil sampling temperature in Celsius.

NOTE 1 Corrected values are valid only for comparing results obtained at different oil temperatures. Actual values of water in oil at sampling points are the measured values, not the corrected ones.

NOTE 2 This formula is not applicable to temperatures below 20 °C.

Acceptable limits for these parameters depend largely upon the type of equipment. However, high values of dielectric dissipation factor, or low values of resistivity, may deleteriously affect the power factor and/or the insulation resistance of the electrical equipment.

These parameters are very sensitive to the presence of soluble polar contaminants, ageing products or colloids in the oil. Changes in the levels of the contaminants can be monitored by measurement of these parameters even when contamination is so slight as to be near the limit of chemical detection.

6.6 Dielectric Dissipation Factor (DDF) and resistivity

The rate of increase of acidity of oil in service is a good indicator of the ageing rate. The acidity level is used as a general guide for determining when the oil should be replaced or reclaimed.

The acidity of a used oil is due to the formation of acidic oxidation products. Acids and other oxidation products will, in conjunction with water and solid contaminants, affect the dielectric and other properties of the oil. Acids have an impact on the degradation of cellulosic materials and may also be responsible for the corrosion of metal parts in a transformer.

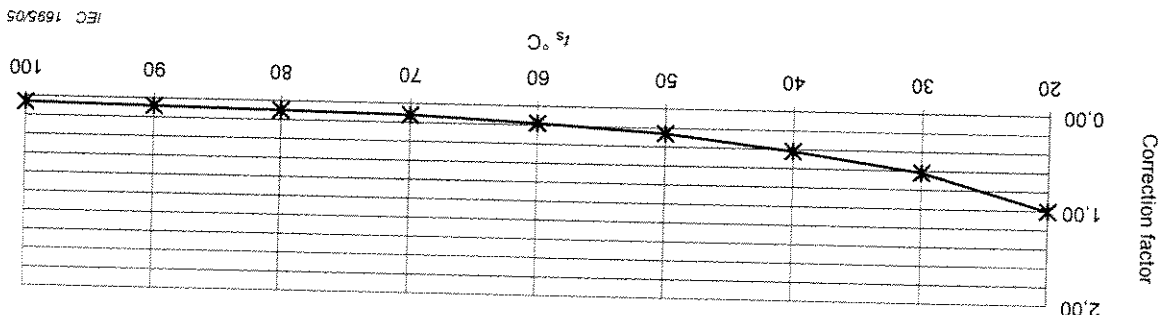
The acidity (neutralization value) of oil is a measure of the acidic constituents or contaminants in the oil.

6.5 Acidity

Measured water content
10 mg/kg
Sampling temperature
40 °C
Correction factor (from formula 2)
0.45
Corrected dissolved water content (10×0.45)
4.5 mg/kg

EXAMPLE

Figure 2 – Typical correction factors



IEC 1695/05

There is generally a relationship between DDF and resistivity, with resistivity decreasing as DDF increases. It is normally not necessary to conduct both tests on the same oil and generally DDF is found to be the more common test. Resistivity and DDF are temperature dependent and Figure 3 illustrates typical changes of resistivity with temperature for insulating oils that are virtually free from solid contamination and water.

Useful additional information can be obtained by measuring resistivity or DDF at both ambient temperature and a higher temperature such as 90 °C.

In the case of VHV and UHV instrument transformers, special attention shall be paid to DDF as it has been reported that a higher value of DDF may lead to thermal runaway leading to transformer failure.

Oils classified as 'good' (see 10.4) will have characteristics similar to curves A and B in Figure 3 and will result in satisfactory test results being obtained at both the higher and lower temperatures.

Oils classified as 'poor' (see 10.4) will have characteristics similar to curve C and will result in a satisfactory test result at 90 °C coupled with an unsatisfactory value at the lower temperature. This is an indication of the presence of water or degradation/deterioration products precipitable in the cold without any significant amount of chemical degradation or general contamination. Unsatisfactory results at both temperatures indicate a greater extent of contamination and that it may not be possible to restore the oil to a satisfactory condition by reconditioning.

The measurement of resistivity is also considered to be of value for monitoring oils in service, as it has been shown to be reasonably proportional to oxidation acids and to be affected by undesirable contaminants such as metal salts and water. Other compounds present in used oils, which can affect resistivity, include aldehydes, ketones and alcohols. An increase in temperature reduces the resistivity, as also does water when precipitated at low temperature due to the saturation point being reached.

NOTE It has been observed in instrument transformers that some types of oil may experience a huge increase in DDF after a very short oxidation time, leading to failure of the equipment. It is therefore recommended to measure the DDF of the unused oil after subjecting it to a short oxidation period according to IEC 61125 method C to verify that the oil is suitable for this application.

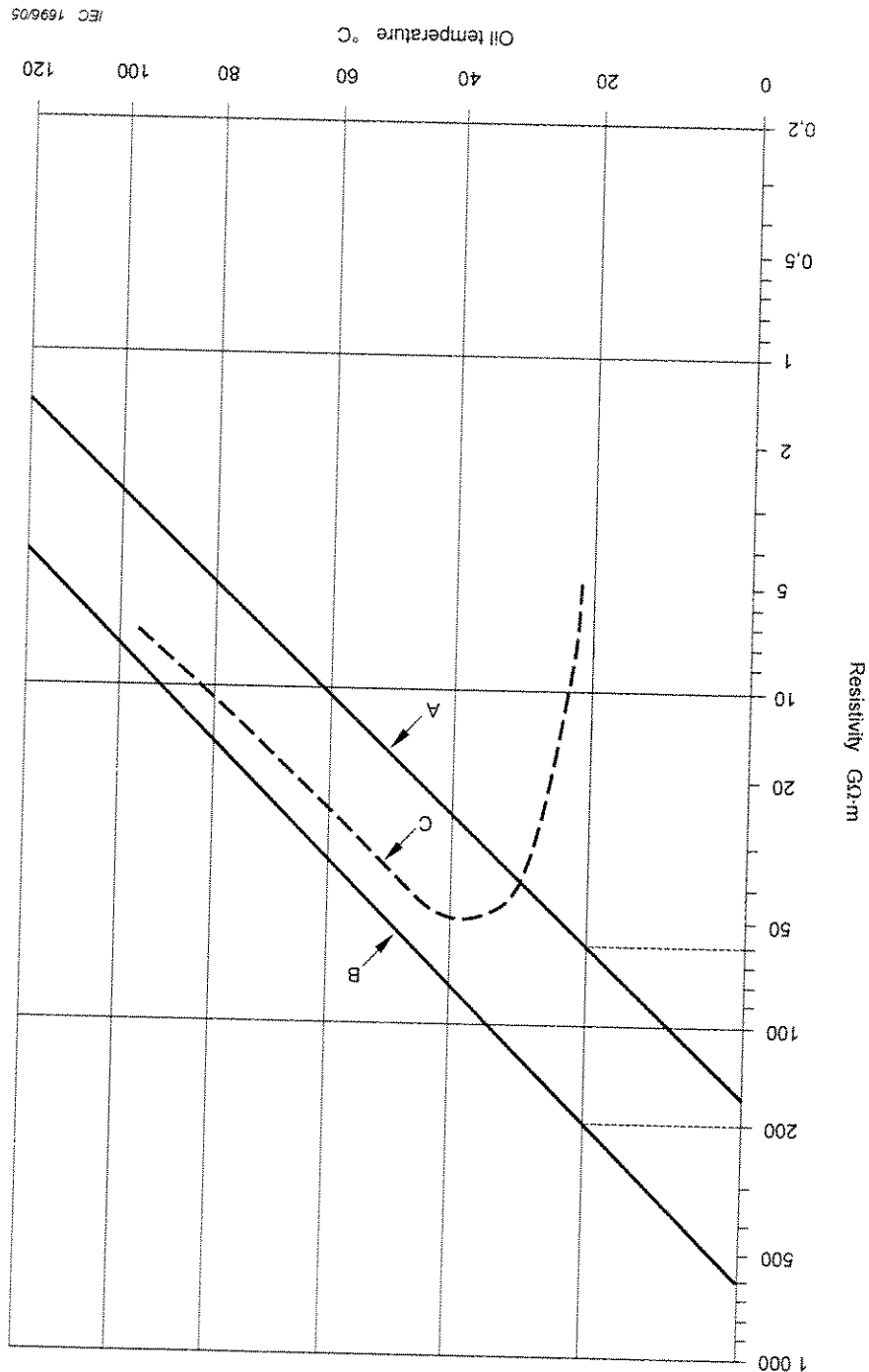


Figure 3 – Example of variation of resistivity with temperature for insulating oils

Line A: Dry oil having a resistivity of 60 GΩ·m at 20 °C
 Line B: Dry oil having a resistivity of 200 GΩ·m at 20 °C
 Line C: Wet oil that is 100 % saturated at 35 °C
 NOTE: In transformers in service, the behaviour of line C is unlikely to occur in the windings, but rather along tank walls or other very cold surfaces.

IEC 1696/05

6.7 Inhibitor content and oxidation stability

6.7.1 Oxidation stability

The ability of mineral electrical insulating oil to withstand oxidation under thermal stress and in the presence of oxygen and a copper catalyst is called oxidation stability. It gives general information about the life expectancy of the oil under service conditions in electrical equipment. The property is defined as resistance to formation of acidic compounds, sludge and compounds influencing the Dielectric Dissipation Factor (DDF) under given conditions. For oils complying with IEC 60296, these conditions are detailed in IEC 61125 method C and the limits of acceptable performance in IEC 60296.

The property depends mainly on the refining process and how it is applied to a given feedstock. Refined mineral oils contain, to a varying degree, natural compounds acting as oxidation inhibitors. These are known as natural inhibitors. Oils containing only natural inhibitors are designated as uninhibited oils.

Synthetic oxidation inhibitors can be added to enhance the oxidation stability. In transformer oils, mainly the phenolic type is used and the common and generally accepted compounds are 2,6-di-tert-butyl-paracresol (DBPC) and 2,6-di-tert-butyl-phenol (DBP). The efficacy of added inhibitors will vary with the chemical composition of the base oil.

6.7.2 Monitoring of uninhibited oils

Oxidation of uninhibited oils is normally monitored by the formation of acidic compounds and oil soluble and insoluble sludge. An increase in DDF and reduction in IFT are also signs of oxidation of insulating oils (see 6.5, 6.6 and 6.9).

6.7.3 Monitoring of inhibited oils

Inhibited oils have a different oxidation pattern compared to uninhibited oils. At the beginning of service life, the synthetic inhibitor is consumed with little formation of oxidation products. This is referred to as the induction period. After the inhibitor is consumed, the oxidation rate is determined mainly by the base oil oxidation stability.

The common and easy way to monitor the inhibitor consumption is to measure the inhibitor concentration according to IEC 60666.

The inhibitor content should be monitored at regular intervals the frequency of which will depend upon operational temperature and load levels. When the concentration of inhibitor is reduced, there are three options, namely:

- re-inhibition of the oil to the original value of inhibitor concentration, if other parameters show a low degree of ageing;
- if the oil is in poorer condition, reclaim and inhibit to original concentration;
- continue to use the oil but at low inhibitor concentrations, typically 0,05 %, increase the frequency of sampling.

6.7.4 Oxidation stability tests

To determine "remaining lifetime" the oxidation stability tests specified in IEC 61125, method C may be used. As these tests are designed for unused oils, interpretation of test results may be difficult. Although not widely used, IEC 61125, method B can be beneficial in determining the remaining induction period for inhibited oils.

6.8 Sediment and sludge

This test distinguishes between sediment and sludge.

Sediment is insoluble material present in the oil.

Sediment includes:

- insoluble oxidation or degradation products of solid or liquid insulating materials;
- solid products arising from the conditions of service of the equipment; carbon, metal, metallic oxides;
- fibres, and other foreign matter, of diverse origins.

Sludge is a polymerized degradation product of solid and liquid insulating material. Sludge is soluble in oil up to a certain limit, depending on the oil solubility characteristics and temperature. At sludge levels above this, the sludge is precipitated, contributing as an additional component to the sediment.

The presence of sediment and/or sludge may change the electrical properties of the oil, and in addition, deposits may hinder heat-exchange, thus encouraging thermal degradation of the insulating materials.

Sediment and sludge should be measured according to the method described in Annex C.

6.9 Interfacial tension (IFT)

The interfacial tension between oil and water provides a means of detecting soluble polar contaminants and products of degradation. This characteristic changes fairly rapidly during the initial stages of ageing but levels off when deterioration is still moderate. A rapid decrease of IFT may also be an indication of compatibility problems between the oil and some transformer materials (varnishes, gaskets), or of an accidental contamination when filling with oil. However, oils with interfacial tension values at or near the lower limit value given in Table 5 should be further investigated.

With overloaded transformers, the deterioration of materials is rapid and IFT is a tool for detection of deterioration.

6.10 Particle count

The presence of particles in insulating oil in electrical equipment may have a number of possible sources. The equipment itself may contain particles from manufacturing and the oil may contain particles from storage and handling if not properly filtered. Wear and the ageing of oil and solid materials may produce particles during the service life of equipment. Localized overheating over 500 °C may form carbon particles. The carbon particles produced in the on-load tap-changer diverter switch may migrate by leakage into the bulk oil compartment to contaminate the oil-immersed parts of the transformer. A typical source of metallic particles is wear of bearings of the pumps.

The effect of suspended particles on the dielectric strength of insulating oil depends on the type of particles (metallic, fibres, sludge, etc.) and on their water content.

Historically, some failures on HV transformers have been associated with particle contamination. Traditional dielectric breakdown voltage tests are not sufficient to identify the problem and particle counting methods have been advised as monitoring tools [3, 4] (Table B.1).

6.11 Flash-point

Breakdown of the oil caused by electrical discharges or prolonged exposure to very high temperatures may produce sufficient quantities of low molecular weight hydrocarbons to cause a lowering of the flash-point of the oil.

A low flash-point may be an indication of the presence of volatile combustible products in the oil. This may result from contamination by a solvent but, in some cases, the cause has been observed to be extensive sparking discharges.

6.12 Compatibility of insulating oils

Oil complying with IEC 60296 and of the same type shall be used for topping up and/or refilling.

Unused oils conforming to IEC 60296 and containing the same or no additives are considered to be compatible with each other and can be mixed in any proportion.

Field experience indicates that problems are not normally encountered when unused oil is added in small percentage, e.g. less than 5 %, to used oils classified as 'good' (see 10.4), though larger additions to heavily aged oil may cause sludge to precipitate.

A compatibility test may be needed to determine the feasibility of mixing unused oils of different origin, with oil in service. For mixing used oils, a compatibility study is strongly recommended. Reference to the oil supplier is recommended if any doubts concerning compatibility arise.

In the compatibility study, as described below, the characteristics of the mixture should not be less favourable than the worst individual oil.

Oils should be mixed in the same proportions as in the application, or if not known in a 50/50 ratio.

The minimum analyses considered necessary for the pure oils and the mixture are

- dielectric dissipation factor,
- acidity,
- pour point,
- oxidation stability according to IEC 61125, Method C, including acidity, sludge and DDF after ageing. Test time should be according to the oil type as stated in IEC 60296.

Experience is very limited regarding the use of oil containing pour-point depressants to top-up naturally low pour-point oils. However, laboratory investigations suggest that no significant deterioration of low temperature behaviour is likely to occur.

Compatibility tests are particularly necessary in the case of oils containing additives. Again, reference to the oil supplier or to the equipment manufacturer is recommended.

6.13 Pour-point

Pour-point is a measure of the ability of the oil to flow at low temperature. There is no evidence to suggest that this property is affected by normal oil deterioration. Changes in pour-point can normally be interpreted as the result of topping-up with a different type of oil.

6.14 Density

Density may be useful for type identification. In cold climates, the density of oil may be important in determining its suitability for use. For example, ice crystals formed from separated water may float on oil of high density and lead to flashover on subsequent melting. However, density is not significant in comparing the quality of different samples of oil. There is no evidence that density is affected by normal oil deterioration.

6.15 Viscosity

Viscosity is an important controlling factor in the dissipation of heat. Ageing and oxidation of the oil tend to increase viscosity. Viscosity is also affected by temperature. Normal ageing and oxidation of the oil will not significantly affect its viscosity. Only under extreme conditions of corona discharges or oxidation may this occur.

6.16 Polychlorinated biphenyls (PCBs)

Polychlorinated biphenyls (PCBs) are a family of synthetic chlorinated aromatic hydrocarbons, which have good thermal and electrical properties. These properties combined with excellent chemical stability made them useful in numerous commercial applications. However, their chemical stability and resistance to biodegradation has given cause for concern in terms of environmental pollution. This increasing concern over the environmental impact of PCBs has progressively restricted their use since the early 1970s and their use in new plants and equipment was banned by international agreement in 1986. Unfortunately, the use of common handling facilities has led to widespread contamination of mineral insulating oil.

The PCB content of oil in new equipment should be measured to confirm that the oil is PCB free. Thereafter, whenever there is a risk of potential contamination (oil treatment, transformer repairs, etc.) the oil should be analysed and if PCB content is found to exceed defined limits, appropriate action should be taken (see 12.4).

NOTE Limits will be as defined by local regulations.

6.17 Corrosive sulphur

Sulphur is present in refined oil as sulphur-containing molecules. The amount depends on oil refining processes, degree of refining and crude oil type.

Due to poor refining or contamination, reactive compounds giving corrosion at normal operating temperatures may be present. They are detected by standard tests as ISO 5662 [14] or DIN 51353 [5], which are included in all major oil specifications.

At high temperatures, >300 °C, sulphur-containing oil molecules may decompose and react with metal surfaces to form metal sulphides. Such reactions may take place in switching equipment and will impact the conductivity of contacts.

7 Sampling of oil from equipment

It is essential that every effort be made to ensure that samples are representative of the insulating oil in equipment. Experience indicates that oil is sometimes rejected unjustifiably because inadequate care has been taken whilst sampling. Careless sampling procedures or contamination in the sample container will lead to erroneous conclusions concerning quality and incur waste of time, effort and expense involved in obtaining, transporting and testing the sample.

Whenever is possible, sampling from equipment shall be carried at normal operating conditions or very shortly after de-energization.

Sampling should be done by an experienced person, who has received adequate training, in accordance with IEC 60475.

Where available, the manufacturer's instructions should be followed.

8 Categories of equipment

In order to take into account different user requirements, equipment has been placed in various categories as shown in Table 2 below.

Table 2 – Categories of equipment

Category	Type of equipment
Category O	Power transformers/reactors with a nominal system voltage of 400 kV and above.
Category A	Also power transformers of any rated voltage above 170 kV and below 400 kV. equipment for special applications operating under onerous conditions.
Category B	Power transformers/reactors with a nominal system voltage above 72,5 kV and up to and including 170 kV (other than those in category A).
Category C	Power transformers/reactors for MV/LV application e.g. nominal system voltages up to and including 72,5 kV and traction transformers (other than those in category A).
Category D	Oil-filled circuit breakers with a nominal system voltage exceeding 72,5 kV.
	Oil-filled switches, a.c. metal-enclosed switchgear and control gear with a nominal system voltage greater than or equal to 16 kV.
Category E	Instrument/protection transformers with a nominal system voltage above 170 kV.
Category F	Diverter tanks of on-load tap-changers, including combined selector/diverter tanks.
Category G	Oil-filled circuit breakers with a nominal system voltage up to and including 72,5 kV.
	Oil-filled switches, a.c. metal-enclosed switchgear and control gear with a nominal system voltage less than 16 kV.

Table 2 (continued)

NOTE 1 Separated selector tanks of on-load tap-changers belong to the same category as the associated transformer.
NOTE 2 Oil-impregnated paper bushings and other hermetically sealed equipment may be placed in category D or E if a routine monitoring programme is desired. The manufacturer's instructions should be referred to.
NOTE 3 Regardless of size or voltage, a risk assessment may justify condition-monitoring techniques usually appropriate to a higher classification.
NOTE 4 For practical and economical reasons, some electrical utilities may decide that their small transformers up to 1 MVA and 36 kV are not included in this classification. Routine monitoring programmes may not be considered economical for this type of equipment. Where a monitoring programme is required for these transformers, the guidelines given for category C should be adequate.

9 Evaluation of mineral insulating oil in new equipment

A substantial proportion of electrical equipment is supplied to the final user already filled with mineral oil. In such cases, as the oil has already come into contact with insulating and other materials, it can no longer be considered as "unused oil" as defined in IEC 60296. Therefore its properties have to be regarded as those applicable to "used oil", even though the electrical equipment itself may not have been energized.

Oil properties should be appropriate to the category and functions of the equipment (Table 3).

The extent of the changes in properties may vary with the type of equipment due to the different types of material and ratios of liquid-to-solid insulation, and should be within the limits of Table 3.

As the characteristics of oil in new equipment are an integral part of that equipment design, the user may request these characteristics to be better than the minimum standards suggested in Table 3, which are based on the experience of many years of operating practice.

**Table 3 – Recommended limits for mineral insulating oils after filling
in new electrical equipment prior to energization**

Property	Highest voltage for equipment kV		
	<72,5	72,5 to 170	>170
Appearance	Clear, free from sediment and suspended matter		
Colour (on scale given in ISO 2049)	Max. 2,0	Max. 2,0	Max. 2,0
Breakdown voltage (kV)	>55	>60	>60
Water content (mg/kg) ^a	20 ^b	<10	<10
Acidity (mg KOH/g)	Max. 0,03	Max. 0,03	Max. 0,03
Dielectric dissipation factor at 90°C and 40 Hz to 60 Hz ^c	Max. 0,015	Max. 0,015	Max. 0,010
Resistivity at 90 °C (GΩm)	Min. 60	Min. 60	Min. 60
Oxidation stability	As specified in IEC 60296		
Interfacial tension (mN/m)	Min. 35	Min. 35	Min. 35
Total PCB content (mg/kg)	Not detectable (< 2 total)		
Particles	-	-	See Table B.1 ^d

Table 3 (continued)

a	The values are not corrected for temperature since not enough time may have elapsed to reach an equilibrium between oil and cellulose insulation.
b	For use in transformers under 72.5 kV class, the maximum water content should be agreed between supplier and user depending upon local circumstances.
c	Higher dielectric dissipation factor values may indicate excessive contamination, or the misapplication of solid materials used in manufacture, and should be investigated.
d	A determination of particle size and quantity should be made as a baseline for future comparison in transformers >170 kV.

10 Evaluation of oil in service

10.1 General

Insulating oil in service is subjected to heat, oxygen, water and other catalysts, all of which are detrimental to the properties of the oil. In order to maintain the quality of the oil in service, regular sampling and analysis should be performed.

Often the first sign of oil deterioration may be obtained by direct observation of the oil clarity and colour through the conservator's visor. From an environmental point of view, this simple and easy inspection can be also used to monitor leakage and spills of oil to the transformer's surrounding.

The interpretation of results, in terms of the functional deterioration of the oil, should be done by experienced personnel based on the following elements of risk management and life cycle analysis:

- characteristic values for the type and family of oil and equipment, developed by statistical methods;
- evaluation of trends and the rate of variation of the values for a given oil property;
- normal, or typical values, for "fair" or "poor" for the appropriate type and family of equipment.

In the case of oil contaminated with PCB, environmental impact is a critical factor to consider, as are local regulations. If it is suspected that oil has become contaminated with PCB, specific analyses should be undertaken and interpretation of the results should be used in risk assessment to take into account prevention and mitigation of potential damage to the environment and to avoid unreasonable risks for staff and the public.

10.2 Frequency of examination of oils in service

It is impossible to lay down a general rule for the frequency of examination of oils in service, which will be applicable to all possible situations that might be encountered. The optimum frequency will depend on the type, function, voltage, power, construction and service conditions of the equipment, as well as the condition of the oil as determined in the previous analysis. A compromise will often have to be found between economic factors and reliability requirements.

Much greater difficulties exist in deciding frequency of testing and permissible oil deterioration levels which are acceptable for all applications of insulating oil in relation to differences in operating policies, reliability requirements and types of electrical system. For example, large power companies may find the full application of these recommendations to distribution transformers uneconomical. Conversely, the industrial user, whose activities depend on the reliability of his power supply, may wish to institute more frequent and stricter controls of oil quality as a means of guarding against power failures.

By way of a guide, a suggested frequency of tests suitable for different types of equipment is given in Table 4. However, some equipment is designed having systems that are designed to control exposure of the oil to atmosphere. Where such systems are maintained in good condition, less frequent testing may be appropriate based on life cycle analysis (LCA) and/or life cycle management (LCM) and risk assessment (RA).

Generally, check measurements should be carried out on the basis of the following criteria, which apply particularly to transformer oils:

- Characteristics may be checked periodically, at intervals as suggested in Table 4, unless otherwise defined.
- The frequency of examination may be increased where any of the significant properties indicates that the oil is in fair or poor condition, or when trend analysis indicates significant changes.
- The oxidation of the oil will accelerate with increased temperature and in the presence of oxygen and water. Therefore heavily loaded transformers may need more frequent oil-sampling and complementary testing such as interfacial tension.
- The testing frequency should be established by means of a cost/benefit evaluation based on life cycle analysis and risk assessment. For some owners this approach may indicate different testing frequencies for distribution transformers (category C) than those indicated in Table 4. For instance, some electrical utilities may prefer not to perform this programme to this type of transformers and small industries may prefer to include this type of transformers even in a higher category.

Table 4 – Recommended frequency of testing⁽¹⁾

Property (2)	Equipment category (3)					
	O	A	B	C (4)	D (5)	E (5)
Group 1 (Routine Tests) (5) – Years	1 – 2	1 – 3	1 – 4	2 – 6	1 – 2	2 – 6
Group 2 (Complementary tests)	Note (7)					
Group 3 (Special investigative tests)	Note (8)					

NOTE 1 These proposed periods refer to a normal routine maintenance programme. Should one or more of the measured properties indicate that the oil is in a fair or poor condition or if an abnormal ageing trend is observed, these periods should be shortened according to the importance of the equipment. These periods may also be shortened in the case of oils contaminated by PCB in order to minimize any potential environmental impact caused by malfunctioning equipment.

NOTE 2 Groups 1, 2 and 3 are defined in Clause 4 and in footnote d) to Table 1.

NOTE 3 Equipment categories are defined in Table 2.

NOTE 4 Subclause 10.2 (bullet 4).

NOTE 5 Equipment categories D and E. After the first sample has been taken, the user, after consultation with the manufacturer and/or laboratory, may decide to lengthen the sampling period.

NOTE 6 Group 1 tests should be performed after filling or refilling the transformer, prior to energizing.

NOTE 7 These tests may be done periodically but less frequently than routine tests. The frequency will depend upon the type of oil, age and equipment. First (benchmark) measurements should be carried out in new or refurbished equipment prior to energization.

NOTE 8 These are very special tests that need be effected only under special circumstances.

10.3 Testing procedures

10.3.1 General

The venue for testing and the number and type of tests that can be carried out on a given sample of oil may vary depending on local circumstances and economic considerations.

Oil in service will vary widely in the extent of degradation and the degree of contamination. In general, a single type of test is insufficient to evaluate the condition of the oil sample.

Evaluation of condition should preferably be based upon the composite evaluation of significant characteristics determined in suitably qualified and properly equipped laboratories. However, some users find it advantageous to carry out field screening tests.

10.3.2 Field tests

Oil tests (see Table 1) are chosen to

- obtain a prompt estimation of oil condition;
- establish the classification of service-aged oils (see 10.4);
- eliminate any changes to the oil sample's properties due to transportation to a laboratory and/or storage of oil samples; and
- the presence of on-site and on-line test instruments with an accuracy comparable to laboratory test instruments.

Some field tests are less accurate than laboratory tests. Field tests are usually limited to visual inspection (colour and appearance), breakdown voltage, water content and, with less accuracy, acidity. These tests may sometimes be used for the assessment of service-aged oils in accordance with 10.4, though, more often, field tests are carried out to identify oil samples requiring laboratory evaluation.

Experience has shown that breakdown voltage and water content tests carried out on site may produce reliable results and may be used for acceptance tests.

10.3.3 Laboratory tests

A complete examination scheme includes all the tests listed in Table 1. However, these tests may be sub-divided into three groups and tests applicable to one or more groups may be required according to the specific requirements (Tables 1 and 3).

10.4 Classification of the condition of oils in service

It is virtually impossible to set hard and fast rules for the evaluation of oil in service or recommend test limits for all possible applications of insulating oil in service. The classification and any consequent corrective action should only be taken after due consideration of the results of all tests. The trend of such results over a period of time is considered essential information when arriving at a final decision.

According to local or current industrial experience, oils in service may be classified as "good", "fair" or "poor" based on the evaluation of significant properties and their ability to be restored to the characteristics desired. Table 5 provides guidance to assist in this classification process.

Good

Oil in normal condition; continue normal sampling.

Fair

Oil deterioration detectable; more frequent sampling recommended (Table 5).

Poor

Oil deterioration abnormal; immediate action advisable.

10.5 Corrective action

In general, two types of contamination/deterioration of the oil can be considered: physical and chemical. Each one requires a different remedial action as described in Table 6.

The following recommendations should also be noted:

a) Where a test result is outside the limits recommended in Table 5, it should be compared with previous results and, if appropriate, a fresh sample obtained for confirmation before any other action is taken.

b) If rapid deterioration or acceleration in the rate of deterioration is observed, more frequent tests (Table 4) should be instituted promptly and appropriate remedial action should be taken. It may be desirable to consult the manufacturer of the equipment.

Table 5 – Application and interpretation of tests

Property	Category (1)	Recommended action limits			Recommended action (2) (3)	Notes
		Good	Fair	Poor		
Colour and appearance	All	Clear and without visible contamination		Dark and/or turbid	As dictated by other tests.	Dark colour is a symptom of chemical contamination or ageing. Turbidity is a symptom of high water content.
Breakdown voltage (kV)	O, A, D	> 60	50 – 60	< 50	Good: continue normal sampling.	
	B, E	> 50	40 – 50	< 40	Fair: more frequent sampling. Check other parameters e.g. water, particle content and perhaps DDF/resistivity and acidity.	
	C	> 40	30 – 40	< 30	Poor: recondition oil (see 12.2) or, alternatively, if more economical because other tests indicate severe ageing, replace the oil.	
	F	Tap changer of neutral end tap changers on O, A, B, C transformers < 25				
	G	Single phase or connected tap changers on O, A, B transformers < 40				
Water content (mgH ₂ O/kg _{oil} at 20 °C) (corrected to an equivalent value at 20 °C)	O, A, D	< 5	5 – 10	> 10	Good: continue normal sampling.	Warning: when oil temperature during sampling is at or above 20 °C, values in mg/kg as measured should always be corrected to 20 °C before comparing them to the corrected limit values of Table 6.
	B, E	< 5	5 – 15	> 15	Fair: more frequent sampling. Check other parameters e.g. breakdown voltage, particle content and perhaps DDF/resistivity and acidity.	When the oil temperature during sampling is less than 20 °C or where no significant amounts of cellulosic insulation are present, refer to Annex A.
	C	< 10	10 – 25	> 25	Poor: check source of water, recondition oil (see 12.2) or, alternatively, if more economical because other tests indicate severe ageing, replace the oil.	
	F	As per appropriate transformer				
	G	Not a routine test				

Table 5 (continued)

Property	Category (1)	Recommended action limits			Recommended action (2) (3)	Notes
		Good	Fair	Poor		
Acidity (mgKOH/g oil)	O, A, D	< 0,10	0,10 – 0,15	> 0,15	Good: continue normal sampling. Fair: more frequent sampling. Check the presence of sediment and sludge.	
	B, E	< 0,10	0,10 – 0,20	> 0,20	Poor: starting from a value of 0,15 the owner may wish to produce a trend analysis based on specific Life Cycle Analysis (LCA) and Risk assessment (RA). A decision may then be made at which point to reclaim the oil (see 12.3) or, alternatively, if more economical and other tests indicate severe ageing, replace the oil.	
	C	< 0,15	0,15 – 0,30	> 0,30		
	F, G	Not a routine test				
	Dielectric dissipation factor at 40 Hz to 60 Hz at 90 °C	O, A	< 0,10	0,10 – 0,20	> 0,20	Good: continue normal sampling.
B, C		< 0,10	0,10 – 0,50	> 0,50	Fair: more frequent sampling. Check other parameters.	
D		< 0,01	0,01 – 0,03	> 0,03	Poor: reclaim oil (see 12.3) or, alternatively, if more economical because other tests indicate severe ageing, replace the oil.	
E		< 0,10	0,10 – 0,30	> 0,30	Refer to manufacturer's best practice.	Catastrophic failures have been reported at 0,04 DDF in VHV and EHV type D transformers.
F, G		Not a routine test			Refer to manufacturer's best practice.	

Table 5 (continued)

Property	Category (1)	Recommended action limits			Recommended action (2) (3)	Notes
		Good	Fair	Poor		
Resistivity (GΩm)	At 20 °C				Good: continue normal sampling. Fair: more frequent sampling. Check other parameters. Poor: reclaim oil (see 12.3) or, alternatively, if more economical because other tests indicate severe ageing, replace the oil.	
	O, A	> 200	20 – 200	< 20		
	B, C	> 60	4 – 60	< 4		
	D	> 800	250 – 800	< 250		
	E	> 60	7 – 60	< 7		
	At 90 °C					Not a routine test
	O, A	> 10	1 – 10	< 1		
	B, C	> 3	0,2 – 3	< 0,2		
	D	> 50	10 – 50	< 10		
	E	> 3	0,4 – 3	< 0,4		
Inhibitor content	All	1) Inhibitor concentration of (40 – 60) % of original value and acidity ≤ 0,06 mgKOH/g _{oil} and IFT ≥ 30 mN/m. 2) Inhibitor concentration < 40 % of original value or acidity > 0,06 mgKOH/g _{oil} and IFT < 30 mN/m.			Re-inhibit to original concentration. Reclaim the oil (see 12.3) and re-inhibit to original concentration, or Continue to use in service but at inhibitor concentrations < 0,05 % increase the frequency of monitoring.	Not routine test Restricted to inhibited oils. Consult oil supplier.
Sediment and sludge	All	No sediment or precipitable sludge. Results below 0,02 % by mass may be neglected.			Where sediment is detected recondition oil (see 12.2). Where precipitable sludge is detected reclaim oil (see 12.3). Alternatively if more economical, or other tests dictate, replace the oil.	Not a routine test. Perform only when acidity and dissipation factor values are near the limits.

Table 5 (continued)

Property	Category (1)	Recommended action limits			Recommended action (2) (3)	Notes
		Good	Fair	Poor		
Interfacial tension (mN/m)	O, A, B, C, D	> 28	22 – 28	< 22	Good: continue normal sampling. Fair: more frequent sampling. Poor: check the presence of sediment and sludge.	Not a routine test. May be done as desired.
Particles (counting and sizing)	O, A, B, C, D, E, F	Not a routine test			Refer to manufacturer's best practice.	Values for particles are based on statistical evaluation. (Table B.1)
Flash point	All	Maximum decrease 10 %			Equipment may require inspection. Investigate.	Not a routine test. May be required when an unusual odour is noticed, when an internal fault has occurred or when a transformer is refilled.
PCB	All	Not a routine test			Equipment may require inspection. Investigate.	In some countries, health and safety legislation may preclude a higher limit
NOTE 1 Category of equipment (Clause 8, Table 2).						
NOTE 2 Ensure sampling was properly done (i.e. sampling valve was properly cleaned before sampling (Clause 7) and transportation and storage periods in the laboratory were short.						
NOTE 3 No action should be taken on the basis of one result and one property. All results should be confirmed by a repeat sample, especially if the result appears abnormal compared to the trend of the previous results obtained.						
NOTE 4 If there is a tendency for particles to increase, then it will be necessary to make a determination of metals and other elements dissolved in the oil (see ASTM D 5185 [1]).						

Table 6 – Corrective actions

Cause of oil change		Symptoms	Corrective actions
Physical	- High water content	- Low breakdown voltage value - High particles content - Turbid (not clear)	- Reconditioning (see 12.2)
	- High colour value - Low IFT - High acidity value - High dissipation factor value - Presence of sediments or sludge, or both		
Chemical	- High colour value - Low IFT - High acidity value - High dissipation factor value - Presence of sediments or sludge, or both	- Low inhibitor content	- Reclaiming (see 12.3) - or change the oil Top up with new inhibitor according to recommendations in Table 5
	- Low inhibitor content		
PCB		PCB contamination detectable	- Refer to local regulations (see 12.4)

NOTE In some cases, if the chemical contamination is extremely high, it may be more economical to replace the oil. A reclaiming viability test (see 12.3.3) is recommended.

No action should be taken on the basis of one result and one property. Repeat samples are recommended where the result appears abnormal compared to the trend of the results previously obtained.

11 Handling and storage

CAUTION Safe drum handling and environmental procedures should be adopted according to local regulations. Special attention shall be paid to avoid cross contamination by PCB.

To ensure satisfactory service, the utmost care in handling the oil is essential. Drums should be clearly marked to indicate whether they are for clean or for dirty oil, and should be reserved for the type indicated. Drums should be stored horizontally and placed in such a position that there is a head of oil on the stopper or plug. They should be stored under cover to minimize the ingress of water and to reduce solar thermal cycling due to exposure to sunlight. The use of plastic sheeting is not recommended unless great care is taken to avoid the drums "sweating" with condensation.

During transportation, drums should be in the vertical position for stability and covered to prevent the ingress of water.

In practice, difficulty may be experienced in maintaining the purity of oil when it is transferred from one vessel to another due to the possibility of introducing contamination. Such practice is not recommended without strict adherence to quality control.

It is recognized that storage of oil in damaged drums is not always satisfactory and the transfer of oil from such containers to electrical equipment should be through a suitable treatment plant to remove water and dissolved gases.

In locations with fixed oil-handling equipment, the pipe-work from the clean oil tanks to the electrical apparatus should be kept clean and free from water. Dehydrating breathers should be regularly inspected and maintained. Where portable oil-handling equipment is used, flexible pipe-work and hand pumps should be carefully inspected to ensure that they are free from dirt and water, and should be flushed with clean oil before use. If the clean oil is from drums, it should have been recently tested, and the filling orifices of the drums should be clean.

Hoses used for clean oil and hoses used for dirty oil should be clearly marked and provided with plugs for sealing the ends when not in use. Hoses shall be resistant to oil, as ordinary rubber contains free sulphur, which is corrosive. If wire braided hoses are used, the hoses shall be cross-banded and properly grounded to prevent the build-up of any static charge. For specific problems, reference should be made to the equipment manufacturer's instructions.

12 Treatment

12.1 General

WARNING The treatment of used oil has to be done with proper care. All countermeasures should be taken to minimize any unreasonable risk to workers, public health and the environment. Experienced and qualified personnel well aware of the health and environmental risks associated should always perform oil treatment, strictly in accordance with local regulations. A full Risk Assessment should always be undertaken before commencing any treatment.

Strict control shall be undertaken in order to avoid cross contamination by PCB.

Strict control shall be undertaken to avoid accidental spills to the environment. Pipes, pumps and hoses shall be carefully inspected for tightness.

As oil treatments are usually carried out under vacuum, special attention shall be paid to avoid emissions to the atmosphere.

Oil treatments produce waste, such as spent filters, oil-contaminated absorbents etc. It is therefore necessary to choose the best available technology to minimize production of waste or spent materials and to dispose of waste strictly according to local regulations.

If the treatment is performed on on-load equipment, strict safety measures shall be taken to avoid risks to the workers. Also, safety measures shall be taken to avoid any damage to the equipment itself.

Due care should be taken when working with hot oil. Workers should use appropriate personal protective equipment according to local regulations and the risk assessment.

The properties of the oil after any treatment should be agreed between service provider and customer.

12.2 Reconditioning

12.2.1 General

Reconditioning is a process that eliminates or reduces physical contamination by means of physical processes (filtration, de-humidification, de-gasification, etc.).

Reconditioning is carried out at the user's site, employing physical means only, to remove contaminants from the oil. However, this process does not always result in oil that conforms to Table 3 of this standard.

Reconditioning reduces the particle and water content of the oil. The process may also remove some dissolved gases and other components such as furanic compounds. New datum levels should be established after such a process.

The physical means that are used for removing water and solids from oil include several types of filtration, centrifuging and vacuum dehydration techniques.

If vacuum treatment is not employed it is advisable to limit the temperature to 30 °C. If vacuum used, the initial boiling point of the oil being treated should not be exceeded, to avoid undue loss of lighter fractions. If this information is not available, it is recommended that the oil should not be vacuum treated at temperatures over 85 °C.

NOTE 1 Processing inhibited mineral oil under vacuum and at elevated temperatures may cause partial loss of oxidation inhibitors. The common inhibitors, 2,6-di-tert-butyl-paracresol and 2,6-di-tert-butyl-phenol, are more volatile than mineral insulating oil. The selectivity for removal of water and air in preference to loss of inhibitor and oil is improved by use of a low processing temperature.

If it is desirable to reduce particles or free water, cold treatment at atmospheric pressure may be appropriate.

Filters deal efficiently with solid impurities, but are generally capable of removing only small quantities of free water. Where relatively large quantities of free water are present, most of it can, and should, be removed before filtration of the oil.

NOTE 2 Equipment used for filtering oils subject to the risk of contamination by carbon (e.g. from tapchangers) should not be used for other oils because of the risk of cross-contamination.

To prevent loss of additives, the conditions that have been found satisfactory for most inhibited mineral oil processing are shown in Table 7.

Table 7 – Conditions for processing inhibited mineral insulating oils

Temperature °C	Minimum pressure Pa
40	8
50	15
60	30
70	80
80	200
85	280

Centrifugal separators are, in general, satisfactory for removing free water from oil and can deal also with any finely divided solid impurities.

If oil is purified whilst hot, its viscosity is reduced and the throughput with certain types of purifier is greater. On the other hand, sludge and free water are more soluble in hot oil than in cold. Particles and free water are, therefore, more effectively removed by cold treatment. Dissolved and suspended water and dissolved gases are effectively removed by hot vacuum treatment.

If the oil contains solid matter, it is advisable to pass it through some type of filter before processing it under vacuum.

12.2.2 Reconditioning equipment

12.2.2.1 Filters

Filtering equipment usually forces oil under pressure through absorbing material such as paper or other filter media. Filters of this type are normally used to remove contaminants in suspension. It should be noted that the nominal micron ratings, commonly used to characterise these filters, are based on gravimetric tests and applying efficiency, based on weight, which takes no regard of particle size. (The filter medium should be capable of removing particles larger than 10 µm although local regulations may prescribe a lower value e.g. 5 µm.) Such equipment does not de-gas the oil.

The ability of a filter to remove water is dependent upon the dryness and quantity of the filter medium. When filtering oil that contains water, the water content of the filter medium rapidly comes into equilibrium with the water content of the oil. A continuous indication of the water content of the outgoing oil is useful to monitor the efficiency of the process.

Care should be taken to ensure that paper filters are of the correct grade so that they do not shed fibres.

During service, filters become contaminated with used oil and solid contaminants, therefore the disposal of filters should be carried out strictly in accordance with local regulations. Special consideration shall be given to filters likely to be contaminated with PCB.

12.2.2.2 Centrifuges

In general, a centrifuge can handle a much greater concentration of contaminants than a conventional filter but cannot remove some of the solid contaminants as completely as a filter. Consequently, the centrifuge is generally found in use for rough bulk cleaning where large amounts of contaminated oil are to be handled.

The output of the centrifuge is often put through a filter for the final clean-up.

12.2.2.3 Vacuum dehydrators

The vacuum dehydrator is an efficient means of reducing the gas and water content of a mineral insulating oil to very low values. (The use of a vacuum dehydrator to remove excessive water from paper insulation systems using oil circulation is not an efficient process. Special techniques may need to be considered.)

There are two types of vacuum dehydration, both function at elevated temperature. In one method, the treatment is accomplished by spraying the oil into a vacuum chamber; in the other, the oil flows in thin layers over a series of baffles inside a vacuum chamber. In both types, the objective is to expose a maximum surface and minimum thickness of oil to the vacuum.

In addition to removing water, vacuum dehydration will de-gas the oil and may remove some of the more volatile acids and some of the 2-furfural.

12.2.3 Application to electrical equipment

12.2.3.1 Direct purification

The oil is passed through a purifier and then stored in suitable clean containers. When the electrical equipment is to be refilled, the oil is passed through the purifier again, and then directly into the equipment. This method can be used for switchgear. It is suitable, too, for smaller transformers, but care is needed to ensure that the core, the windings, the interior of the tank and other oil-containing compartments are thoroughly cleaned. The oil-containing compartments of all equipment should also be well cleaned, by means of oil from the purifier.

12.2.3.2 Purification by circulation

The oil is circulated through the purifier, being taken from the bottom of the tank of the electrical equipment and re-delivered to the top. The return delivery should be made smoothly and horizontally at or near the top oil level to avoid, as far as possible, mixing cleaned oil with oil that has not yet passed through the purifier. The circulation method is particularly useful for removing suspended contaminants, but not all adhering contaminants will necessarily be removed.

Experience has shown that it is generally necessary to pass the total volume of oil through the purifier not less than three times, and equipment having an appropriate capacity should be chosen with this in mind. The final number of cycles will depend on the degree of contamination, and it is essential that the process be continued until a sample taken from the bottom of the electrical equipment after the oil has been allowed to settle for a few hours, passes the breakdown voltage test.

It is recommended that the circulation should be performed with the electrical equipment disconnected from the power source. In all cases, the oil should be allowed to stand for some time in accordance with the manufacturer's instructions before the equipment is re-energized.

WARNING It is the practice in some countries to perform this process with the transformer energized, but this shall only be done after full risk assessment has been carried out.

Another technique is sometimes used for transformers, in which oil is continuously circulated during normal service through an adsorbent, such as molecular sieve, thus keeping both oil and windings dry and removing many oil oxidation products. This is a specialized method not further considered in this standard.

12.2.3.3 Sealed instrument transformers

In order to avoid the risk of introducing air into the transformer, which may lead to premature failure, oil reconditioning shall be done strictly in accordance with the manufacturer's instructions pertinent at the time of reconditioning.

12.3 Reclaiming

12.3.1 General

This is a process that eliminates or reduces soluble and insoluble polar contaminants from the oil by chemical and physical processing. Reclamation processes require special competence, equipment and experience. The resulting product should be evaluated on critical parameters to achieve information about process efficiency and to be able to estimate remaining lifetime. This process may result in oil, which originally conformed to IEC 60296, being restored to an acceptable standard. Reclamation of oils of moderate to high acidity will usually result in oils with a lower oxidation resistance than the original new oil.

Before performing a reclamation process, a laboratory feasibility test is recommended.

There are two types of oil reclaiming: percolation and contact.

12.3.2 Reclaiming by percolation

The full process consists of three consecutive steps:

- 1) The oil, being taken from the bottom of the electrical equipment, is heated to a given temperature and circulated through a filter (to eliminate the particles and suspended solids) being re-delivered to the top.
- 2) It is then circulated through one or more cartridges containing fuller's earth or other suitable material, to eliminate soluble polar contaminants.
- 3) The oil is finally circulated through a reconditioning device (vacuum dehydrator or centrifuge) to eliminate water and gases.

Fuller's earth is an active material containing both internal and external polar active sites, which allow the non-polar components of the oil to pass through without retention but which retains the polar contaminants or degradation compounds dissolved in the oil.

Several different clays are available that have proven suitable for these purposes. The most widely used are of the sepiolite, bentonite, attapulgite or montmorillonite type of which fuller's earth is the most commonly used. They are constituted of silicate anions $[\text{Si}_2\text{O}_5]_n$ condensed with octahedral layers of the type $\text{X}(\text{OH})_2$ where X may be magnesium, aluminium, etc.

Normally, fuller's earth is treated to increase its specific surface area and the concentration and polarity of its Lewis acids. Fuller's earth can be used alone or in combination with other chemicals like trisodium phosphate, activated charcoal and sodium silicate.

The retention of contaminants by adsorbent active sites is, generally, improved by temperature, thus the process normally takes place at 60 °C to 80 °C.

Experience has shown that it is usually necessary to pass the total volume of oil through the adsorbent not less than three times, and equipment of appropriate capacity should be chosen for this purpose. The final number of cycles will depend on the degree of initial contamination and the desired final level for properties.

In the case of highly contaminated equipment, it is usual to transfer all the oil to a suitable clean container, reclaim a small portion of the oil and use it to thoroughly wash the electrical equipment, especially the windings. This portion of the oil is disposed of according to local regulations, and the remaining oil is then reclaimed as described above.

It is important to bear in mind that a small portion of the oil, less than 5 %, remains absorbed by the adsorbent, thus some unused oil shall be used for topping-up the equipment at the end of the process.

During service, the adsorbent becomes contaminated with used oil and solid contaminants, therefore the disposal, or re-activation, of the substance should be carried out strictly in accordance with local regulations. Special consideration will have to be given to adsorbents likely to be contaminated with PCB.

12.3.3 Reclaiming by contact

This process consists of stirring the contaminated oil, in the presence of fuller's earth, in a suitable container. It is not an appropriate system for industrial applications as it needs very long outage periods for the electrical equipment, but may be useful for the recycling of large amounts of waste oils.

Normally this process is used in the laboratory to investigate the feasibility of a reclamation process in a given oil and to estimate the final levels of the properties that can be reached by reclamation in the field.

12.3.4 Renewal of additives

As oil reclamation is performed after the oil is aged, it is inevitable that the natural inhibitors in the oil have been spent. It is therefore recommended that the additives be replaced in the reclaimed oil after the reclaiming process and before the equipment is re-energized. The most widely used additives are 2,6-di-tert-butyl-p-cresol (DBPC) and 2,6-di-tert-butyl-phenol (DBP).

12.4 Decontamination of oils containing PCB

12.4.1 General

Transformers containing PCB contaminated oils shall not be considered waste whilst in service. Should the oil become accidentally contaminated, there are several processes and techniques available for either on-site and off-site decontamination of PCB contaminated oils. These processes are based on chemical reactions between PCBs and the reagent to remove the chlorine present. All PCB decontamination methods, either off-site or on-site, shall be applied by skilled companies complying fully with local regulations.

Off-site decontamination techniques are limited by considerations for the safe transportation of contaminated equipment and liquid to an authorized oil processing facility and are the subject of local regulations.

12.4.2 Dehalogenation processes using sodium and lithium derivatives

These processes are typically applied in batch and use reagents based on metallic sodium, sodium hydride, lithium hydride and additives, for the dehalogenation of PCB in the oil. This type of process is typically run under pressure and medium to high temperature (150 °C – 300 °C). This temperature is higher than the flash point of the oil (140 °C – 150 °C) and therefore introduces subsequent safety risks.

WARNING Proper measures shall be taken to minimize the risk of fire or explosion, especially in the presence of wet oil.

12.4.3 Dehalogenation processes using polyethyleneglycol and potassium hydroxide (KPEG)

This process, developed to overcome the problems associated with the use of metallic sodium, uses a liquid reagent based on polyethyleneglycol (PEG) and an alkaline metal hydroxide such as potassium hydroxide (KOH). This type of process, which is run at temperatures of 130 °C – 150 °C, has a limited efficiency on some types of contaminants (e.g. Aroclor 1242).

12.4.4 Dehalogenation in continuous mode by closed circuit process

This process uses a solid reagent consisting of a high molecular weight glycol mixture, a mixture of bases and a radical promoter or other catalyst for chemical conversion of organic chlorine to inert salts, on a high surface area particulate support.

This process normally runs at 80 °C – 100 °C and has the capability to decontaminate equipment on-site, through continuous circulation of the oil in a closed system (without draining the oil or using auxiliary tanks), using the solvent capability of the oil for continuous extraction of PCB from solid materials inside the equipment.

13 Replacement of oil in electrical equipment

13.1 Replacement of oil in transformers rated below 72,5 kV and in switchgear and associated equipment

A small extra quantity of oil is needed to rinse the interior of the tank and the immersed parts. It is essential that the tank and the surfaces of conductors and insulators be cleaned effectively and subsequently be kept free from fibres. Such fibres are readily introduced by the use of unsatisfactory cleaning materials during plant maintenance; in practice the only efficient and permissible materials are synthetic. It is also essential that the tank and other surfaces be kept free from water.

A final pressure flush with clean oil of known quality has proved beneficial in the removal of fibres and other extraneous material.

Application of a vacuum extraction procedure has proved valuable in removing other contaminants, provided the equipment can withstand the vacuum.

There should be as little aeration as possible during the filling of tanks and, as far as possible, the end of the delivery pipe should be held below the surface of the oil in order to avoid splashing; alternatively, the tanks should be filled from the bottom.

There should be a standing period of not less than 12 h to allow de-aeration before commissioning transformers (1 h may be adequate for switchgear and 1 h – 2 h for power transformers both with a nominal system voltage of less than 16 kV).

13.2 Replacement of oil in transformers rated 72,5 kV and above

Reference should be made to the equipment manufacturer.

13.3 Replacement of oil in electrical equipment contaminated with PCB

Reference should be made to the equipment manufacturer and local regulations. A full risk assessment should be carried out.

Annex A (informative) Sampling temperature below 20 °C

When sampled oil temperatures are below 20 °C, a different approach for the interpretation of analytical results is needed.

The water solubility of a typical, unused mineral insulating oil is given by the formula:

$$\text{Log } W_s = 7,0895 - \left(\frac{1567}{T} \right) \quad (3)$$

where W_s is the solubility of water in unused mineral oil in mg/kg and T is the absolute temperature of the oil in Kelvin.

And

$$\% \text{ Saturation} = (\text{mg/kg of water}/W_s) \times 100$$

Example (for an unused oil):

Water content = 10 mg/kg

Oil temperature = 20 °C

$$\text{Log } W_s = (-1567/293) + 7,0895 = 1,7414$$

$$W_s = 55,132$$

$$\% \text{ Saturation} = (10/55,132) \times 100 = 18 \%$$

For an oil containing polar compounds from oxidation or contamination, the percentage saturation may be measured by a suitable method such as that in BS 6522 [2] or measured directly with a suitable instrument.

By way of a guide, the condition of cellulosic insulation referred to oil percent saturation is given in Table A.1.

Table A.1 – Guidelines for interpreting data expressed in percent saturation

Percent saturation water in oil	Condition of cellulosic insulation
0 % – 5 %	Dry insulation.
6 % – 20 %	Moderately wet, low numbers indicate fairly dry to moderate levels of water in the insulation. Values toward the upper limit indicate moderately wet insulation.
21 % – 30 %	Wet insulation.
>30 %	Extremely wet insulation.

Source : IEEE C57.106:2002 [6].

Where oil is not in contact with significant amounts of cellulosic insulation (e.g. independent selector or diverter tanks; switchgear), transport of water between liquid and solid insulation will not occur and the correction factors should not be used. Instead, the relative saturation of water in oil should be calculated by the ratio of measured water content to the solubility of water at the average temperature of the oil in the tank of the equipment.

Annex B (informative)

Particles

Table B.1 – Examples of contamination levels (particles) encountered on power transformer insulating oil as measured by laser counting (ISO 4406)

SOURCE: CIGRÉ WG12.17 Brochure 157 – June 2000

(ISO 4406 class)	Maximum count per 100 ml		Contamination designation	Notes
	5 µm	15 µm		
Up to 8/5	250	32	Nil	IEC cleanliness requirement for sample bottles filled with clean solvent
9/6 to 10/7	1 000	130	Low	Excellent oil cleanliness encountered during factory acceptance test and transformer commissioning (Table 4)
11/8 to 15/12	32 000	4 000	Normal	Contamination level typical for transformers in service
16/13 to 17/14	130 000	16 000	Marginal	Contamination level found on a significant number of transformers in service
18/15 and above			High	Contamination level rare and usually indicative of abnormal operating conditions

Considering the field measurements reported by various countries, it is recommended that the contamination level encountered in service should be classified as indicated in this table. A more refined classification would not be realistic in view of the variations that can occur in this type of measurement.

Annex C (informative)

Test method for determination of sediment and sludge

C.1 Sediment determination

To determine sediment:

- weigh a P 10 glass filter [12] or a suitable 8 μm membrane filter to the nearest 0,1 mg;
- filter 100 g of an homogenized sample of the oil through the glass filter;
- flush the filter with 100 ml of n-heptane in portions until the filter is free from oil;
- dry the filter at 105 °C, weigh the filter and calculate the amount of sediment as a percentage.

C.2 Sludge determination

To determine sludge:

- take 25 g of the filtered oil from the sediment determination;
- follow the procedure for sludge precipitation and calculation described in 1.9.1 of IEC 61125.

Bibliography

- [1] ASTM D 5185, Standard test method for determination of additive elements, wear metals, and contaminants in used lubricating oils and determination of selected elements in base oils by inductively coupled plasma atomic emission spectrometry (ICP-AES)
- [2] BS 6522, Method for determination of percentage water saturation of insulating oil
- [3] CIGRE, Technical Brochure 157:2000, Effect of particles on transformer dielectric strength
- [4] CIGRE, Technical Brochure 227:2003, Life management techniques for power transformer
- [5] DIN 51353, Testing of insulating oils; Detection of corrosive sulphur; Silver strip test
- [6] IEEE C57.106:2002, Guide for acceptance and maintenance of insulating oil in equipment
- [7] IEC 60354, Loading guide for oil-immersed power transformers²
- [8] IEC 60567, Guide for the sampling of gases and of oil from oil-filled electrical equipment and for the analysis of free and dissolved gases
- [9] IEC 60599, Mineral oil-impregnated electrical equipment in service – Guide to the interpretation of dissolved and free gases analysis
- [10] IEC 61198, Mineral insulating oils – Methods for the determination of 2-furfural and related compounds
- [11] ISO 4406, Hydraulic fluid power – Fluids – Method for coding the level of contamination by solid particles
- [12] ISO 4407, Hydraulic fluid power – Fluid contamination – Determination of particulate contamination by the counting method using an optical microscope
- [13] ISO 4793, Laboratory sintered (fritted) filters – Porosity grading, classification and designation
- [14] ISO 5662, Petroleum products – Electrical insulating oils – Detection of corrosive sulphur³
- [15] ISO/IEC Guide 73, Risk management – Vocabulary – Guidelines for use in standards

² IEC 60354 is due to be re-issued as IEC 60076-10.
³ This publication has been withdrawn